LAPORAN PRAKTIKUM

POSTTEST (4)

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

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KELAS (C'1)

PROGRAM STUDI INFORMATIKA

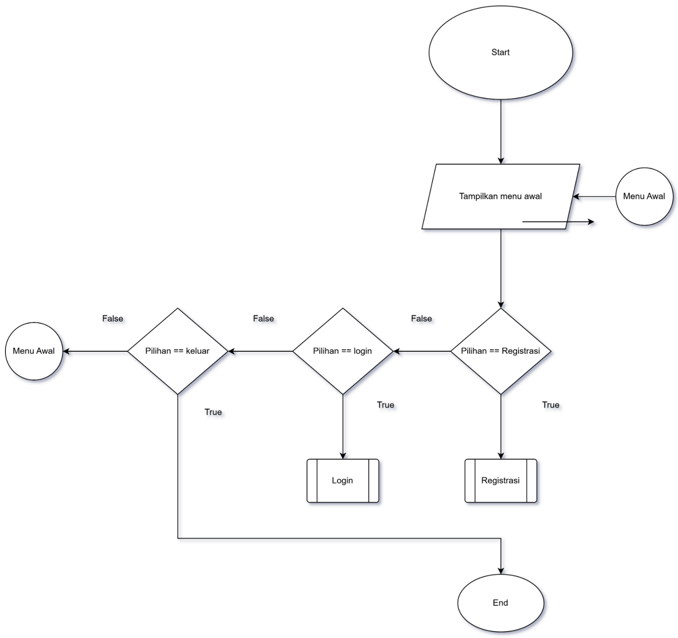
UNIVERSITAS MULAWARMAN

SAMARINDA

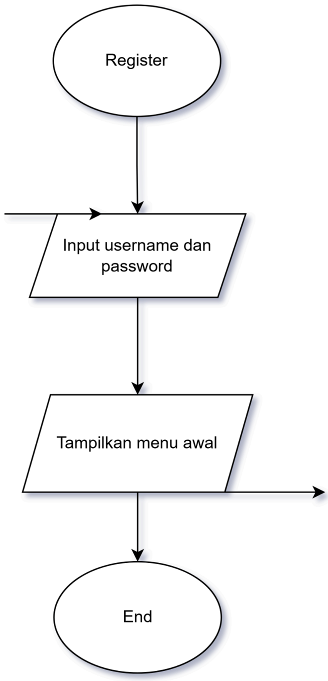
2025

1. Flowchart

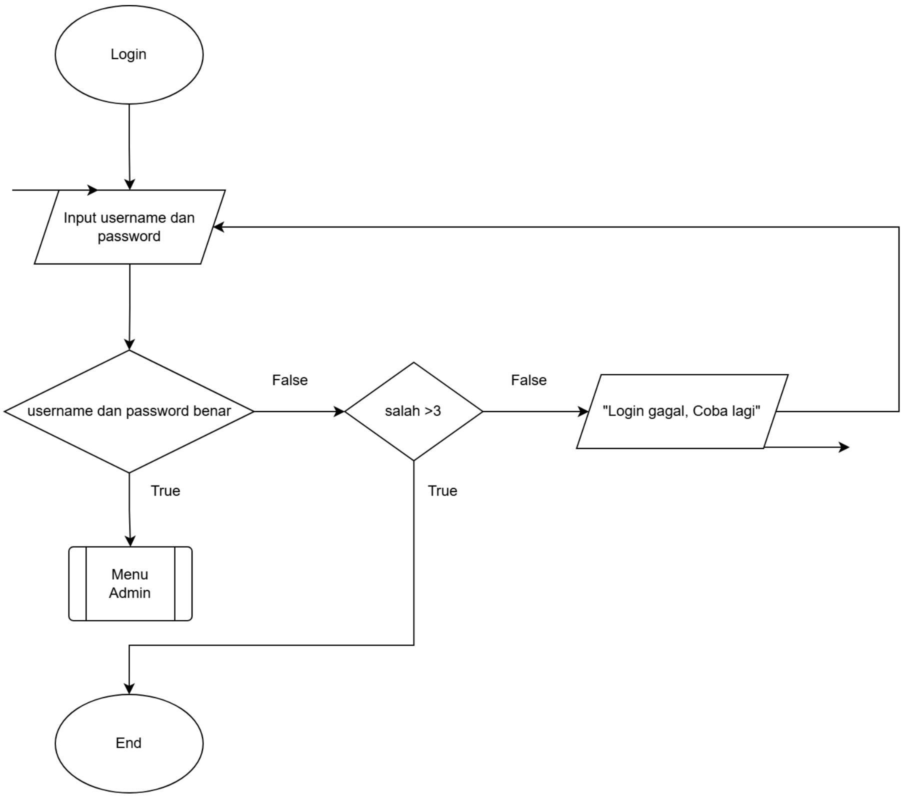
Flowchart ini membantu memvisualisasikan keputusan bercabang (pakai simbol belah ketupat) dan tindakan proses (pakai persegi panjang), sehingga mudah melihat jalur mana yang ditempuh sesuai kondisi.. Kemudian flowchart menggunakan predefined proses agar menunjukkan bahwa ada proses/fungsi yang terdefinisikan



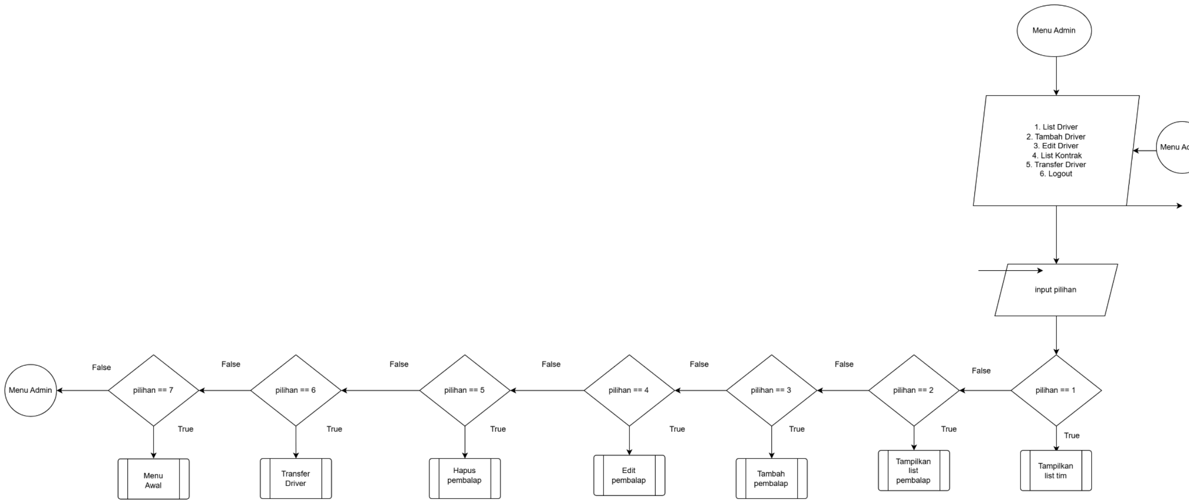
Gambar 1.1 Flowchart CRUD Menu awal



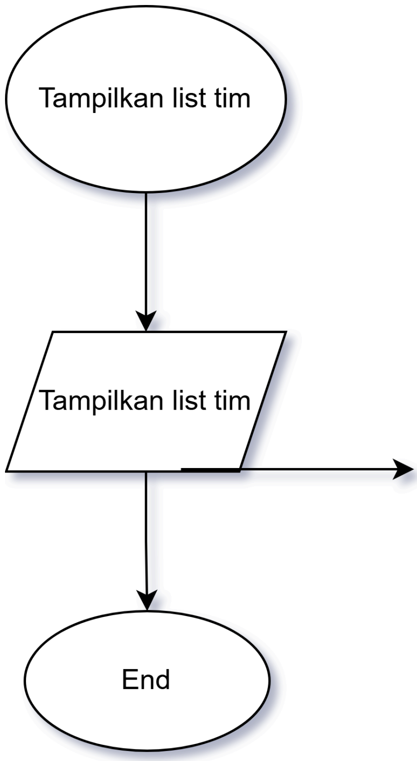
Gambar 1.2 Flowchart Register



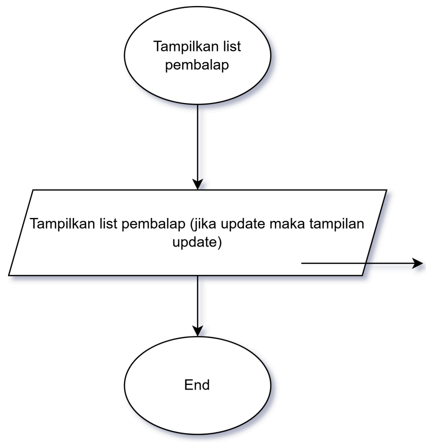
Gambar 1.3 Flowchart Login



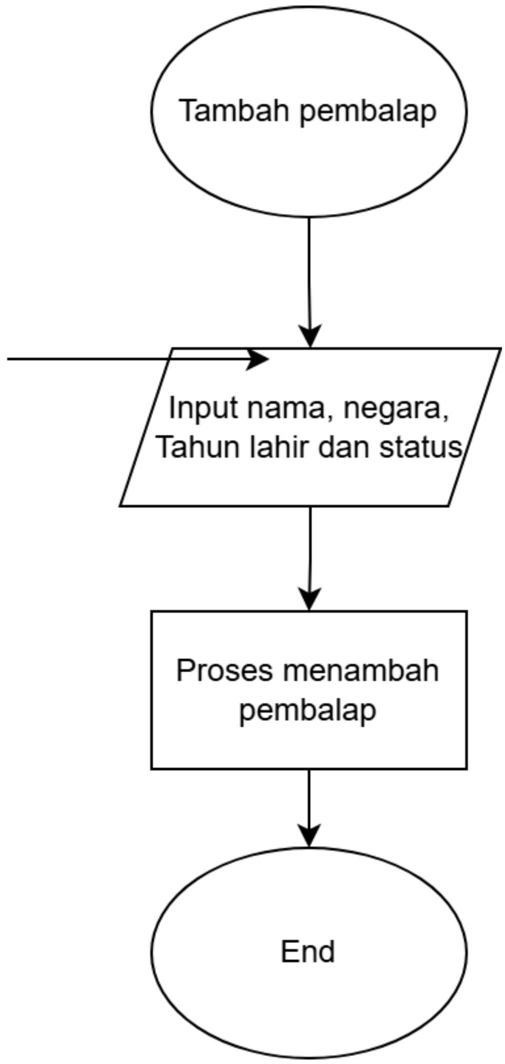
Gambar 1.4 Flowchart Menu Admin



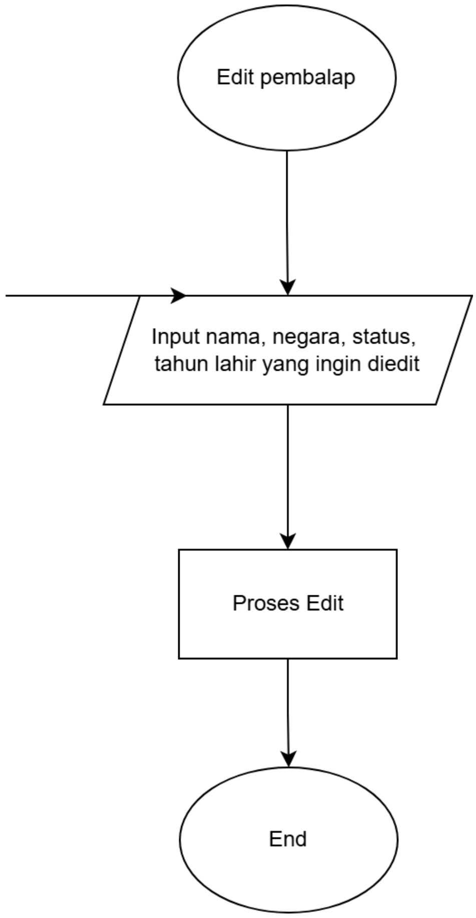
Gambar 1.5 Tampilkan List Tim



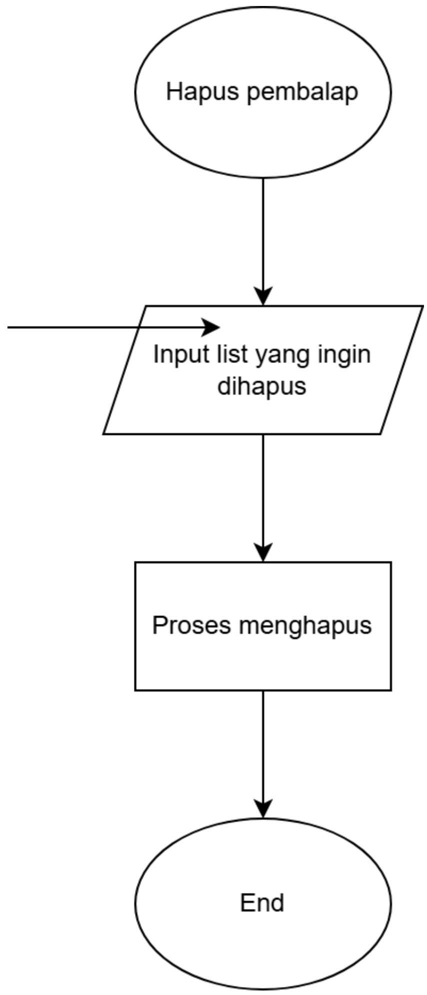
Gambar 1.6 Tampilkan list pembalap



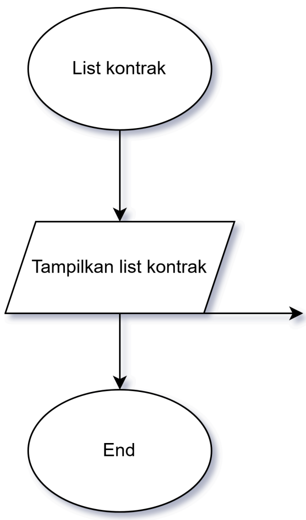
Gambar 1.7 Tambah pembalap



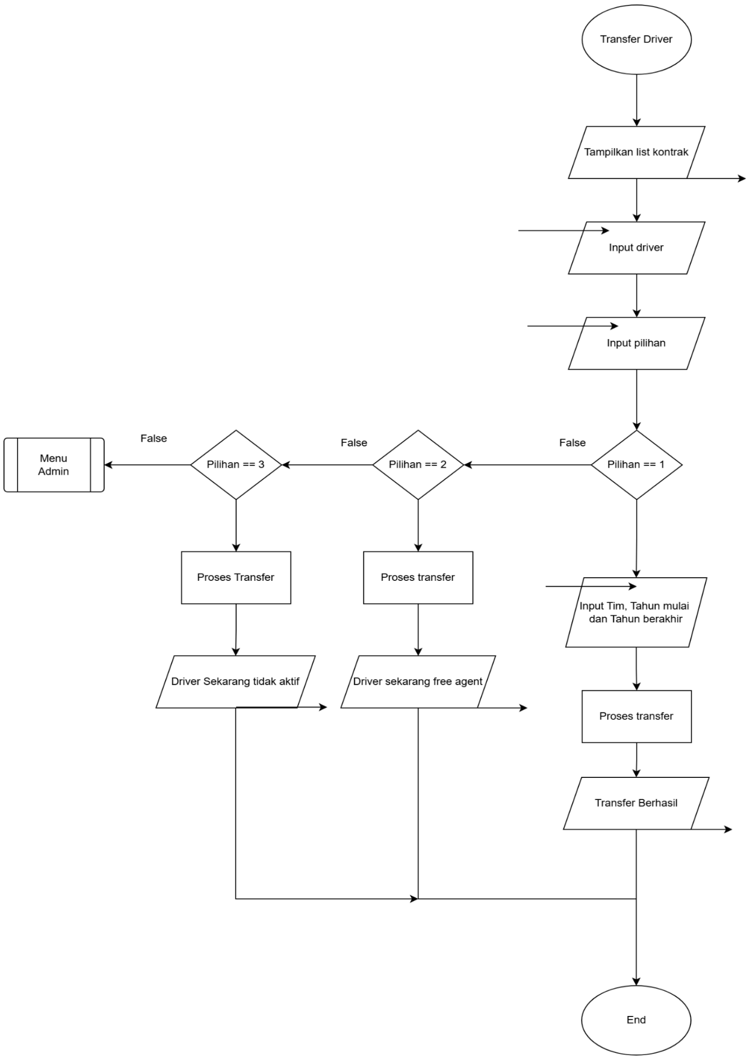
Gambar 1.8 Edit Pembalap



Gambar 1.9 Hapus pembalap



Gambar 1.10 List Kontrak



Gambar 1.11 Transfer Driver

2. Deskripsi Singkat Program

Secara keseluruhan, program ini berfungsi sebagai sistem manajemen kontrak dan informasi pembalap F1 yang sederhana, dengan kontrol akses berdasarkan role dan menu interaktif di konsol.

3. Source Code

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;
```

```
const int MAX_DRIVERS = 50;
const int MAX_USERS = 20;
const int MAX_TEAMS = 11;
const int MAX_CONTRACTS = 100;
const int CURRENT_YEAR = 2026;

struct Driver {
    string name;
    string nationality;
    int birthYear;
    string status;
};

struct User {
    string username;
    string password;
};

struct Team {
    string name;
};

struct Contract {
    int driverIndex;
    int teamIndex;
    int startYear;
    int endYear;
};

void initDriverF1(Driver *d, int *count) {
    string nama[] = {
        "Charles Leclerc","Lewis Hamilton","George Russell","Kimi
Antonelli",
        "Lando Norris","Oscar Piastri","Fernando Alonso","Lance Stroll",
        "Esteban Ocon","Oliver Bearman","Max Verstappen","Sergio Perez",
        "Arvid Lindblad","Pierre Gasly","Valtteri Bottas","Isack Hadjar",
        "Nico Hulkenberg","Gabriel Bortoletto","Liam Lawson","Alexander
Albon",
        "Carlos Sainz","Franco Colapinto"
    };
```

```
    string negara[] = {
        "Monaco","UK","UK","Italy",
        "UK","Australia","Spain","Canada",
        "France","UK","Netherlands","Mexico",
        "Sweden","France","Finland","France",
        "Germany","Brazil","New Zealand","Thailand",
        "Spain","Argentina"
    };

    int tahun[] = {
        1997,1985,1998,2006,
        1999,2001,1981,1998,
        1996,2005,1997,1990,
        2004,1996,1989,2003,
        1987,2004,2002,1996,
        1994,2003
    };

    for (int i = 0; i < 22; i++) {
        d[i] = {nama[i], negara[i], tahun[i], "Aktif"};
    }

    *count = 22;
}

void initTeams(Team *t, int *count) {
    string nama[] = {
        "Scuderia Ferrari HP","Mercedes AMG Petronas","McLaren Mastercard",
        "Oracle Red Bull Racing","Visa Cash App Racing Bulls","BWT Alpine",
        "Aston Martin Aramco","Audi Revolut","Atlassian Williams",
        "Cadillac","HAAS TGR"
    };

    for (int i = 0; i < 11; i++) {
        t[i].name = nama[i];
    }

    *count = 11;
}
```

```
void initDefaultUser(User *u, int *count) {
    u[0] = {"Attol","2509106103"};
    *count = 1;
}

void initKontrakDefault(Contract *c, int *cc) {
    int teamMap[22] = {
        0,0,1,1,2,2,6,6,10,10,3,9,4,5,9,3,7,7,4,8,8,5
    };

    for (int i = 0; i < 22; i++) {
        c[i] = {i, teamMap[i], 2025, 2027};
    }

    *cc = 22;
}

void registerUser(User *u, int *count) {
    if (*count >= MAX_USERS) {
        cout << "Jumlah user sudah penuh!\n";
        return;
    }

    cout << "Username: ";
    getline(cin, u[*count].username);
    cout << "Password: ";
    getline(cin, u[*count].password);
    (*count)++;

    cout << "Registrasi berhasil!\n";
}

bool loginUser(User *u, int count) {
    string us, pw;
    cout << "Username: ";
    getline(cin, us);
    cout << "Password: ";
    getline(cin, pw);

    for (int i = 0; i < count; i++) {
        if (u[i].username == us && u[i].password == pw) {
```

```
            return true;
        }
    }

    cout << "Login gagal!\n";
    return false;
}

string getTeamName(int idx, Contract *c, int cc, Team *t, Driver *d) {
    if (d[idx].status == "Free Agent") return "Free Agent";
    if (d[idx].status == "Tidak Aktif") return "-";

    for (int j = 0; j < cc; j++) {
        if (c[j].driverIndex == idx && c[j].endYear >= CURRENT_YEAR) {
            return t[c[j].teamIndex].name;
        }
    }

    return "Free Agent";
}

void listPembalap(Driver *d, int dc, Contract *c, int cc, Team *t) {
    cout << "\n";
    cout << left << setw(4) << "No"
         << setw(25) << "Nama"
         << setw(15) << "Negara"
         << setw(8) << "Umur"
         << setw(15) << "Status"
         << setw(28) << "Tim" << endl;

    cout << string(95, '=') << endl;

    for (int i = 0; i < dc; i++) {
        cout << setw(4) << i + 1
             << setw(25) << d[i].name
             << setw(15) << d[i].nationality
             << setw(8) << CURRENT_YEAR - d[i].birthYear
             << setw(15) << d[i].status
             << setw(28) << getTeamName(i, c, cc, t, d)
             << endl;
    }
```

```
}

void listKontrak(Contract *c, int cc, Driver *d, Team *t) {
    cout << "\n=== LIST KONTRAK ===\n";
    for (int i = 0; i < cc; i++) {
        cout << i + 1 << ". "
             << d[c[i].driverIndex].name << " - "
             << t[c[i].teamIndex].name
             << " (" << c[i].startYear << "-" << c[i].endYear << ")\n";
    }
}

void hapusKontrakDriver(Contract *c, int *cc, int driverIdx) {
    for (int i = 0; i < *cc; i++) {
        if (c[i].driverIndex == driverIdx) {
            for (int j = i; j < *cc - 1; j++) {
                c[j] = c[j + 1];
            }
            (*cc)--;
            i--;
        }
    }
}

void tambahPembalap(Driver *d, int *dc) {
    if (*dc >= MAX_DRIVERS) {
        cout << "Data driver penuh!\n";
        return;
    }

    cout << "Nama: ";
    getline(cin, d[*dc].name);

    cout << "Negara: ";
    getline(cin, d[*dc].nationality);

    cout << "Tahun lahir: ";
    cin >> d[*dc].birthYear;
    cin.ignore();

    cout << "Status (Aktif/Tidak Aktif/Free Agent): ";
```

```
    getline(cin, d[*dc].status);

    (*dc)++;
    cout << "Driver berhasil ditambahkan!\n";
}

void editPembalap(Driver *d, int dc, Contract *c, int cc, Team *t) {
    listPembalap(d, dc, c, cc, t);

    int i;
    cout << "\nPilih driver yang ingin diedit: ";
    cin >> i;
    cin.ignore();

    if (i < 1 || i > dc) {
        cout << "Pilihan tidak valid!\n";
        return;
    }

    i--;

    cout << "Nama baru: ";
    getline(cin, d[i].name);

    cout << "Negara baru: ";
    getline(cin, d[i].nationality);

    cout << "Tahun lahir baru: ";
    cin >> d[i].birthYear;
    cin.ignore();

    cout << "Status baru (Aktif/Tidak Aktif/Free Agent): ";
    getline(cin, d[i].status);

    if (d[i].status == "Free Agent" || d[i].status == "Tidak Aktif") {
        hapusKontrakDriver(c, &cc, i);
    }

    cout << "Data driver berhasil diupdate!\n";
}
```

```
void hapusPembalap(Driver *d, int *dc, Contract *c, int *cc, Team *t) {
    listPembalap(d, *dc, c, *cc, t);

    int idx;
    cout << "\nPilih driver yang ingin dihapus: ";
    cin >> idx;
    cin.ignore();

    if (idx < 1 || idx > *dc) {
        cout << "Pilihan tidak valid!\n";
        return;
    }

    idx--;

    hapusKontrakDriver(c, cc, idx);

    for (int i = idx; i < *dc - 1; i++) {
        d[i] = d[i + 1];
    }
    (*dc)--;

    for (int i = 0; i < *cc; i++) {
        if (c[i].driverIndex > idx) {
            c[i].driverIndex--;
        }
    }

    cout << "Driver berhasil dihapus!\n";
}

void transferDriverKeTim(Contract *c, int *cc, int driverIdx, int teamIdx,
int startYear, int endYear) {
    for (int i = 0; i < *cc; i++) {
        if (c[i].driverIndex == driverIdx && c[i].endYear >= CURRENT_YEAR) {
            for (int j = i; j < *cc - 1; j++) {
                c[j] = c[j + 1];
            }
            (*cc)--;
            i--;
        }
```

```
    }

    c[*cc] = {driverIdx, teamIdx, startYear, endYear};
    (*cc)++;
}

void transferDriver(Driver *d, int dc, Team *t, int tc, Contract *c, int
*cc) {
    int dr, pilihanStatus;

    cout << "\n=== LIST DRIVER ===\n";
    listPembalap(d, dc, c, *cc, t);

    cout << "\nPilih driver: ";
    cin >> dr;

    if (dr < 1 || dr > dc) {
        cin.ignore();
        cout << "Pilihan driver tidak valid!\n";
        return;
    }

    dr--;

    cout << "\n=== PILIH STATUS DRIVER ===\n";
    cout << "1. Aktif (transfer ke tim)\n";
    cout << "2. Free Agent\n";
    cout << "3. Tidak Aktif\n";
    cout << "Pilih: ";
    cin >> pilihanStatus;

    if (pilihanStatus == 1) {
        int tm, startYear, endYear;

        cout << "\n=== LIST TIM ===\n";
        for (int i = 0; i < tc; i++) {
            cout << i + 1 << ". " << t[i].name << endl;
        }

        cout << "\nPilih tim baru: ";
        cin >> tm;
```

```
        if (tm < 1 || tm > tc) {
            cin.ignore();
            cout << "Pilihan tim tidak valid!\n";
            return;
        }

        cout << "Masukkan tahun mulai kontrak: ";
        cin >> startYear;

        cout << "Masukkan tahun selesai kontrak: ";
        cin >> endYear;
        cin.ignore();

        if (endYear < startYear) {
            cout << "Tahun selesai tidak boleh lebih kecil dari tahun
mulai!\n";
            return;
        }

        transferDriverKeTim(c, cc, dr, tm - 1, startYear, endYear);
        d[dr].status = "Aktif";

        cout << "Transfer berhasil!\n";
        cout << "Kontrak baru: " << startYear << "-" << endYear << endl;
    }
    else if (pilihanStatus == 2) {
        cin.ignore();
        hapusKontrakDriver(c, cc, dr);
        d[dr].status = "Free Agent";
        cout << "Driver sekarang berstatus Free Agent!\n";
    }
    else if (pilihanStatus == 3) {
        cin.ignore();
        hapusKontrakDriver(c, cc, dr);
        d[dr].status = "Tidak Aktif";
        cout << "Driver sekarang berstatus Tidak Aktif!\n";
    }
    else {
        cin.ignore();
        cout << "Pilihan status tidak valid!\n";
```

```
    }
}

void menu(Driver *d, int *dc, Team *t, int tc, Contract *c, int *cc) {
    int p;
    do {
        cout << "\n=== MENU ===\n";
        cout << "1. List Driver\n";
        cout << "2. Tambah Driver\n";
        cout << "3. Edit Driver\n";
        cout << "4. Hapus Driver\n";
        cout << "5. List Kontrak\n";
        cout << "6. Transfer Driver\n";
        cout << "7. Logout\n";
        cout << "Pilih: ";

        cin >> p;
        cin.ignore();

        switch (p) {
            case 1:
                listPembalap(d, *dc, c, *cc, t);
                break;
            case 2:
                tambahPembalap(d, dc);
                break;
            case 3:
                editPembalap(d, *dc, c, *cc, t);
                break;
            case 4:
                hapusPembalap(d, dc, c, cc, t);
                break;
            case 5:
                listKontrak(c, *cc, d, t);
                break;
            case 6:
                transferDriver(d, *dc, t, tc, c, cc);
                break;
            case 7:
                cout << "Logout berhasil.\n";
                break;
```

```
            default:
                cout << "Menu tidak valid!\n";
        }
    } while (p != 7);
}

int main() {
    Driver d[MAX_DRIVERS];
    User u[MAX_USERS];
    Team t[MAX_TEAMS];
    Contract c[MAX_CONTRACTS];

    int dc = 0, uc = 0, tc = 0, cc = 0;

    initDefaultUser(u, &uc);
    initDriverF1(d, &dc);
    initTeams(t, &tc);
    initKontrakDefault(c, &cc);

    int p;
    do {
        cout << "\n1. Register\n2. Login\n3. Keluar\nPilih: ";
        cin >> p;
        cin.ignore();

        if (p == 1) {
            registerUser(u, &uc);
        }
        else if (p == 2) {
            if (loginUser(u, uc)) {
                menu(d, &dc, t, tc, c, &cc);
            }
        }
        else if (p == 3) {
            cout << "Program selesai.\n";
        }
        else {
            cout << "Pilihan tidak valid!\n";
        }

    } while (p != 3);
```

```
    return 0;
}
```

4. Hasil Output

```
+-------------------------------------------+
|      SELAMAT DATANG DI BURSA TRANSFER F1  |
+-------------------------------------------+

==================================================
MENU AWAL
==================================================
+-----------------------------------+
| 1.  Register Admin                |
| 2.  Login Admin                   |
| 3.  Keluar                        |
+-----------------------------------+
Pilihan : 2
Username : Attol
Password : 2509106103
Login berhasil.
```

Gambar 4.1 Hasil Output Menu Awal

```
=== MENU ===
1. List Driver
2. Tambah Driver
3. Edit Driver
4. Hapus Driver
5. List Kontrak
6. Transfer Driver
7. Logout
Pilih:
```

Gambar 4.2 Hasil Output Login

```
===================================================================================
DATA PEMBALAP LENGKAP
===================================================================================
+----+--------------------------+----------------+-------------+-------+-------------+
| No | Nama                     | Negara         | Tahun Lahir | Umur  | Status      |
+----+--------------------------+----------------+-------------+-------+-------------+
| 1  | Charles Leclerc          | Monaco         | 1997        | 29    | Aktif       |
| 2  | Lewis Hamilton           | UK             | 1985        | 41    | Aktif       |
| 3  | George Russell           | UK             | 1998        | 28    | Aktif       |
| 4  | Kimi Antonelli           | Italy          | 2006        | 20    | Aktif       |
| 5  | Lando Norris             | UK             | 1999        | 27    | Aktif       |
| 6  | Oscar Piastri            | Australia      | 2001        | 25    | Aktif       |
| 7  | Fernando Alonso          | Spain          | 1981        | 45    | Aktif       |
| 8  | Lance Stroll             | Canada         | 1998        | 28    | Aktif       |
| 9  | Esteban Ocon             | France         | 1996        | 30    | Aktif       |
| 10 | Oliver Bearman           | UK             | 2005        | 21    | Aktif       |
| 11 | Isack Hadjar             | France         | 2003        | 23    | Aktif       |
| 12 | Max Verstappen           | Netherlands    | 1997        | 29    | Aktif       |
| 13 | Carlos Sainz Jr.         | Spain          | 1994        | 32    | Aktif       |
| 14 | Alexander Albon          | UK             | 1996        | 30    | Aktif       |
| 15 | Valtteri Bottas          | Finland        | 1989        | 37    | Aktif       |
| 16 | Sergio Perez             | Mexico         | 1990        | 36    | Aktif       |
| 17 | Nico Hulkenberg          | Germany        | 1987        | 39    | Aktif       |
| 18 | Gabriel Bortoleto        | Brazil         | 2004        | 22    | Aktif       |
| 19 | Pierre Gasly             | France         | 1996        | 30    | Aktif       |
| 20 | Franco Colapinto         | Argentina      | 1999        | 27    | Aktif       |
| 21 | Liam Lawson              | New Zealand    | 2002        | 24    | Aktif       |
| 22 | Arvid Lindblad           | Sweden         | 2004        | 22    | Aktif       |
+----+--------------------------+----------------+-------------+-------+-------------+
```

Gambar 4.3 Hasil Output List pembalap

```
=== MENU ===
1. List Driver
2. Tambah Driver
3. Edit Driver
4. Hapus Driver
5. List Kontrak
6. Transfer Driver
7. Logout
Pilih: 2
Nama: Benjamin Seskowi
Negara: Solovenia
Tahun lahir: 2002
Status (Aktif/Tidak Aktif/Free Agent): Aktif
Driver berhasil ditambahkan!
```

Gambar 4.4 Output Create Pembalap

```
 *  Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe
==========================================================================================
1   Charles Leclerc         Indonesia     29      Aktif         Scuderia Ferrari HP
2   Lewis Hamilton          UK            41      Aktif         Scuderia Ferrari HP
3   George Russell          UK            28      Aktif         Mercedes AMG Petronas
4   Kimi Antonelli          Italy         20      Aktif         Mercedes AMG Petronas
5   Lando Norris            UK            27      Aktif         Mercedes AMG Petronas
6   Oscar Piastri           Australia     25      Aktif         McLaren Mastercard
7   Fernando Alonso         Spain         45      Aktif         Aston Martin Aramco
8   Lance Stroll            Canada        28      Aktif         Aston Martin Aramco
9   Esteban Ocon            France        30      Aktif         HAAS TGR
10  Oliver Bearman          UK            21      Aktif         HAAS TGR
11  Max Verstappen          Netherlands   29      Aktif         Oracle Red Bull Racing
12  Sergio Perez            Mexico        36      Aktif         Cadillac
13  Arvid Lindblad          Sweden        22      Aktif         Visa Cash App Racing Bulls
14  Pierre Gasly            France        30      Aktif         BWT Alpine
15  Valtteri Bottas         Finland       37      Aktif         Cadillac
16  Isack Hadjar            France        23      Aktif         Oracle Red Bull Racing
17  Nico Hulkenberg         Germany       39      Aktif         Audi Revolut
18  Gabriel Bortoletto      Brazil        22      Aktif         Audi Revolut
19  Liam Lawson             New Zealand   24      Aktif         Visa Cash App Racing Bulls
20  Alexander Albon         Thailand      30      Aktif         Atlassian Williams
21  Carlos Sainz            Spain         32      Aktif         Atlassian Williams
22  Franco Colapinto        Argentina     23      Aktif         BWT Alpine

Pilih driver: 5

=== PILIH STATUS DRIVER ===
1. Aktif (transfer ke tim)
2. Free Agent
3. Tidak Aktif
Pilih: 1

=== LIST TIM ===
1. Scuderia Ferrari HP
2. Mercedes AMG Petronas
3. McLaren Mastercard
4. Oracle Red Bull Racing
5. Visa Cash App Racing Bulls
6. BWT Alpine
7. Aston Martin Aramco
8. Audi Revolut
9. Atlassian Williams
10. Cadillac
11. HAAS TGR

Pilih tim baru: 2
Masukkan tahun mulai kontrak: 2028
Masukkan tahun selesai kontrak: 2035
Transfer berhasil!
Kontrak baru: 2028-2035
```

Gambar 4.5 CRUD Transfer Driver

```
=================================================================================
DATA PEMBALAP LENGKAP
=================================================================================
+----+-------------------------+----------------+-------------+-------+------------+
| No | Nama                    | Negara         | Tahun Lahir | Umur  | Status     |
+----+-------------------------+----------------+-------------+-------+------------+
| 1  | Charles Leclerc         | Monaco         | 1997        | 29    | Aktif      |
| 2  | Lewis Hamilton          | UK             | 1985        | 41    | Aktif      |
| 3  | George Russell          | UK             | 1998        | 28    | Aktif      |
| 4  | Kimi Antonelli          | Italy          | 2006        | 20    | Aktif      |
| 5  | Lando Norris            | UK             | 1999        | 27    | Aktif      |
| 6  | Oscar Piastri           | Australia      | 2001        | 25    | Aktif      |
| 7  | Fernando Alonso         | Spain          | 1981        | 45    | Aktif      |
| 8  | Lance Stroll            | Canada         | 1998        | 28    | Aktif      |
| 9  | Esteban Ocon            | France         | 1996        | 30    | Aktif      |
| 10 | Oliver Bearman          | UK             | 2005        | 21    | Aktif      |
| 11 | Isack Hadjar            | France         | 2003        | 23    | Aktif      |
| 12 | Max Verstappen          | Netherlands    | 1997        | 29    | Aktif      |
| 13 | Carlos Sainz Jr.        | Spain          | 1994        | 32    | Aktif      |
| 14 | Alexander Albon         | UK             | 1996        | 30    | Aktif      |
| 15 | Valtteri Bottas         | Finland        | 1989        | 37    | Aktif      |
| 16 | Sergio Perez            | Mexico         | 1990        | 36    | Aktif      |
| 17 | Nico Hulkenberg         | Germany        | 1987        | 39    | Aktif      |
| 18 | Gabriel Bortoleto       | Brazil         | 2004        | 22    | Aktif      |
| 19 | Pierre Gasly            | France         | 1996        | 30    | Aktif      |
| 20 | Franco Colapinto        | Argentina      | 1999        | 27    | Aktif      |
| 21 | Liam Lawson             | New Zealand    | 2002        | 24    | Aktif      |
| 22 | Arvid Lindblad          | Sweden         | 2004        | 22    | Aktif      |
| 23 | Benjamin Seskowi        | Solovenia      | 2002        | 24    | Tidak Aktif |
+----+-------------------------+----------------+-------------+-------+------------+
Pilih nomor pembalap yang ingin dihapus : 23
Pembalap berhasil dihapus.
```

4.6 Output Delete Pembalap

```
| 22 | Arvid Lindblad          | Sweden         | 2004        | 22    | Aktif      |
+----+-------------------------+----------------+-------------+-------+------------+
Pilih nomor pembalap yang ingin diedit : 22
Nama baru        : Arvid Limbad
Negara baru      : Swedia
Tahun lahir baru : 2007
Pilih status baru:
1. Aktif
2. Tidak Aktif
Pilihan          : 1
Pembalap berhasil diedit.
```

Gambar 4.7 Edit Pembalap

5. Langkah-langkah GIT

```
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git add .
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git commit -m "KOMAT KAMIT"
[main aede32e] KOMAT KAMIT
 3 files changed, 233 insertions(+)
 create mode 100644 post-test/post-test-7/2509106103-KhayzanDwitraAttala-PT-7 (1).pdf
 create mode 100644 post-test/post-test-7/2509106103-PT-7.drawio.png
 create mode 100644 post-test/post-test-7/2509106103_KhayzanDwitraAttala-PT-7.py
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 868.02 KiB | 27.13 MiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Khayzannn/praktikum-apd.git
   609b85f..aede32e  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7>
```

Gambar 5.1 Langkah - Langkah Git

5.1 GIT Add

Git add adalah perintah diGit yang digunakan untuk menambahkan perubahan pada berkas di direktori kerja ke area staging (atau indeks), mempersiapkannya untuk dimasukkan ke dalam commit berikutnya.

5.2 GIT Commit

Git commit adalah perintah di sistem kontrol versi Git untuk menyimpan (melakukan "commit") serangkaian perubahan yang telah dipilih (di-staging) ke dalam repositori lokal Anda, membuat sebuah "snapshot" atau titik pemeriksaan pada riwayat proyek dengan deskripsi perubahan.

5.3 GIT Push

Git push adalah perintah dalam Git yang digunakan untuk mengunggah (mengirimkan) perubahan yang telah dicatat di repositori lokal Anda ke repositori jarak jauh (seperti GitHub), sehingga memperbarui cabang tersebut dan membagikan pekerjaan dengan tim atau menyinkronkan dengan versi pusat.